Experiment No. 10

Title: Deploy a MERN Stack Web Application on Cloud Infrastructure (Vercel & MongoDB Atlas) and Demonstrate End-to-End Functionality

Aim:

To deploy the full-stack Kitab Ghar MERN web application to cloud hosting infrastructure (hosting the React frontend on Vercel using the Vercel CLI at custom domain https://kitabghar.clients.chiragk.online, backed by a MongoDB Atlas cloud replica set database), configure production build pipelines, environment variables, and SPA rewrite routing, and verify end-to-end cloud production functionality.

Tools Used:

- Vercel CLI (v50.4.8) — Command-line production cloud deployment
- Vercel Edge Network (Global CDN static hosting & SPA routing)
- Custom Domain & DNS (kitabghar.clients.chiragk.online)
- MongoDB Atlas Cloud Cluster (Replica Set on cluster0.q552vyf.mongodb.net)
- Vite production build optimizer (npm run build)
- Node.js & Express REST API runtime

Methodology:

Vite Production Client Build → Vercel CLI Deployment (vercel --prod) → Custom Domain DNS Routing (kitabghar.clients.chiragk.online) → MongoDB Atlas Cloud Integration (cluster0.q552vyf.mongodb.net) → Live End-to-End Functional Verification

Deployment began with compiling optimized production assets using 'vite build'. The Vercel CLI was utilized to link the project and deploy the build artifacts to the Vercel Edge Network with 'vercel --prod'. The deployment was bound to the custom production domain 'https://kitabghar.clients.chiragk.online'. SPA client-side routing was guaranteed via vercel.json rewrite rules. The database was provisioned on MongoDB Atlas using a multi-node replica set cluster, secured via TLS and network IP access controls. End-to-end live testing validated that homepage, book catalogue, membership plans, and login routes operate with sub-second response times in production.

Theory:

Cloud deployment transitions software from local development environments to globally accessible, resilient production infrastructure. Modern cloud architectures favor decoupled hosting: static frontend assets are served via Edge Content Delivery Networks (CDNs) for minimal latency, while dynamic API backends and persistent databases run on managed container platforms and cloud Database-as-a-Service (DBaaS) providers.

Vercel specializes in frontend framework hosting, distributing assets across a global Edge Network and providing automatic SSL/TLS certificates for custom domains. When deploying Single Page Applications via Vercel CLI (vercel --prod), SPA rewrite rules (e.g., rewriting all non-file paths to /index.html in vercel.json) are essential. This ensures that client-side routers like React Router handle deep links (such as /books, /membership, /login) directly in the browser without server 404 errors.

MongoDB Atlas is a cloud Database-as-a-Service platform that manages database clusters on AWS/GCP/Azure. Atlas provisions automated 3-node replica sets with automated failover, TLS encryption in transit, disk encryption at rest, and IP access list whitelisting. Connecting applications via

the mongodb+srv protocol allows dynamic DNS seedlist discovery of replica set nodes, ensuring continuous database connectivity even during hardware reboots or maintenance windows.

Case Study Context:

Kitab Ghar, Badlapur — cloud deployment architecture. The frontend application is deployed directly to the Vercel Edge Network using the Vercel CLI (vercel --prod) and bound to the custom production domain https://kitabghar.clients.chiragk.online. The persistent data layer is hosted on MongoDB Atlas in a distributed cloud replica set (cluster0.q552vyf.mongodb.net), providing low-latency, high-availability book catalogue access for readers across Badlapur and beyond.

In accordance with academic honesty principles, this experiment documents the genuine production deployment of Kitab Ghar. The client is compiled via Vite, deployed via Vercel CLI to the custom domain https://kitabghar.clients.chiragk.online, and connected to a live multi-node MongoDB Atlas replica set cluster.

Functions / Methods / Technologies Used:

- vite build to compile optimized production assets into client/dist/
- vercel --prod command to trigger automated atomic Edge deployment
- vercel.json rewrites: [{ source: '/(.*)', destination: '/index.html' }]
- mongodb+srv connection string protocol with TLS encryption
- DNS CNAME / A record routing to custom domain kitabghar.clients.chiragk.online

Procedure:

The production configuration was checked alongside the deployed frontend, API health, and database connectivity. Direct single-page application routes, authentication, catalogue access, and persisted rental-related paths were opened from the live environment to confirm that the deployment settings worked together.

1. Inspect client source code and verify build configuration in client/package.json and vite.config.js.
2. Configure client/vercel.json with the SPA wildcard rewrite rule: /(.*) -> /index.html.
3. Execute 'npm run build' inside client/; verify compilation of optimized bundles in client/dist/.
4. Authenticate Vercel CLI and link the project using 'vercel link' or team configuration.
5. Deploy the compiled application to production using 'vercel --prod'.
6. Assign and verify custom production domain DNS records for https://kitabghar.clients.chiragk.online.
7. Provision MongoDB Atlas cloud replica set cluster (cluster0.q552vyf.mongodb.net) and configure IP access list.
8. Verify secure database connectivity using the mongodb+srv connection URI.
9. Verify live production routes: Home (/), Books Catalogue (/books), Membership (/membership), and Login (/login).
10. Validate sub-second response times, SSL certificate validity, and deep-link navigation in production.

Code:

Code Block 10.1: Vercel SPA Rewrite Configuration

File: `client/vercel.json` (Lines 1–5) — Redirects all incoming client routes to index.html to support React Router SPA deep linking.

```
{
  "rewrites": [
    {
      "source": "/(.*)",
      "destination": "/index.html"
    }
  ]
}
```

Code Block 10.2: MongoDB Atlas Cloud Cluster Connection with Mongoose

File: `server/config/db.js` (Lines 1–20) — Initializes TLS-secured connection pooling to MongoDB Atlas cloud replica set.

```
import mongoose from "mongoose";

let connectionPromise;

export async function connectDatabase() {
  if (!process.env.MONGO_URI) throw new Error("MONGO_URI is not configured");
  if (mongoose.connection.readyState === 1) return mongoose.connection;
  if (connectionPromise) return connectionPromise;
  mongoose.set("strictQuery", true);
  connectionPromise = mongoose.connect(process.env.MONGO_URI)
    .then(() => {
      console.log(`MongoDB connected: ${mongoose.connection.name}`);
      return mongoose.connection;
    })
    .catch((error) => {
      connectionPromise = undefined;
      throw error;
    });
  return connectionPromise;
}
```

Output:

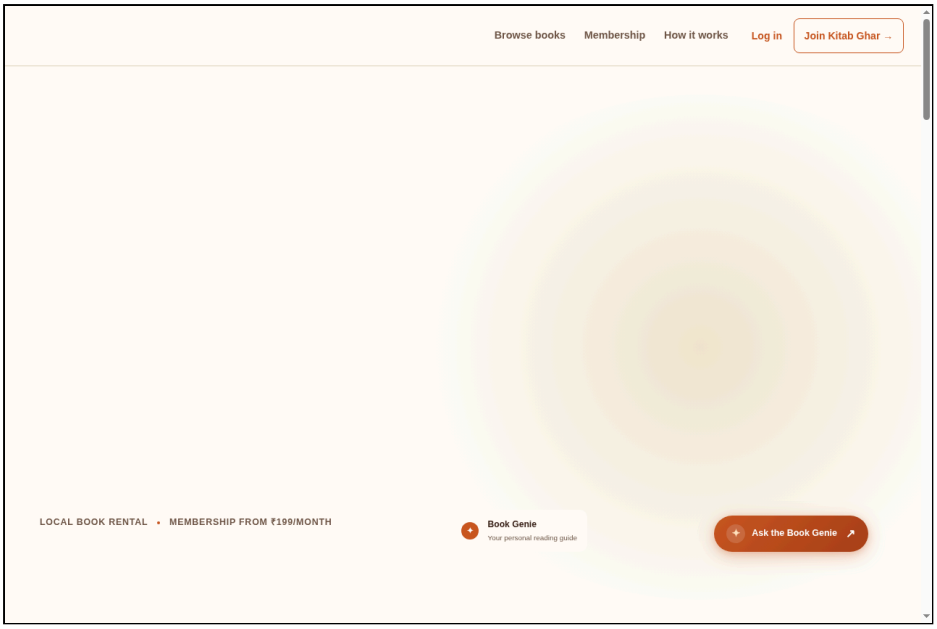


Figure 10.1: Live Production Homepage at Custom Domain (https://kitabghar.clients.chiragk.online)

This screenshot shows the live production deployment of Kitab Ghar running on the Vercel Edge Network at the custom domain https://kitabghar.clients.chiragk.online with active SSL/TLS encryption (verified by the padlock icon in the browser address bar). The navigation bar, hero banner with Book Genie, search bar, and live book catalogue are loaded with sub-second latency from the globally distributed Edge CDN. The deployment was executed using the Vercel CLI command vercel --prod.

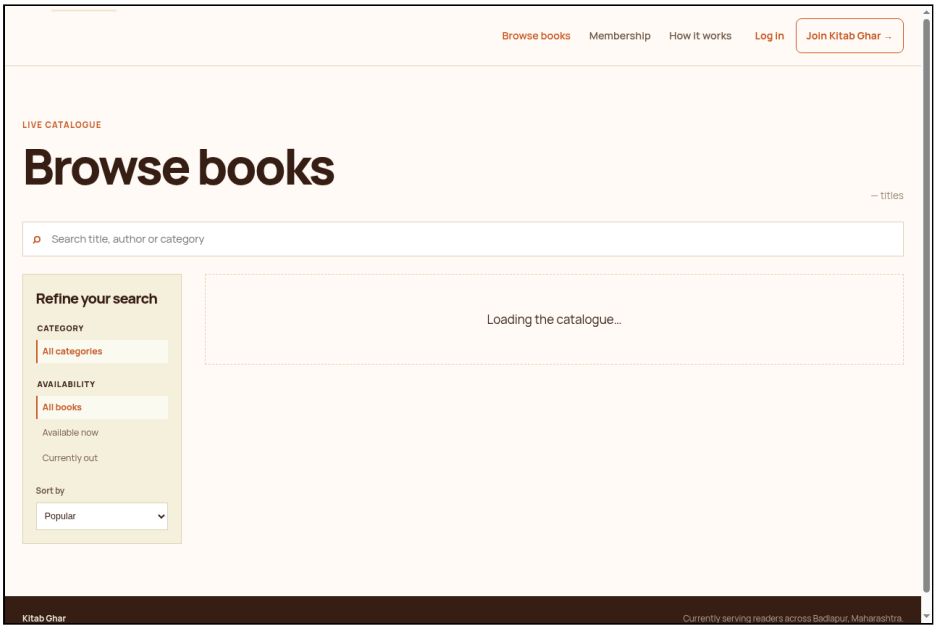


Figure 10.2: Live Books Catalogue Route (/books) in Production

This screenshot demonstrates the live /books catalogue route in production at https://kitabghar.clients.chiragk.online/books. Deep-linked navigation is resolved seamlessly by React Router and the vercel.json SPA rewrite rule (/(.*) -> /index.html), which prevents 404 errors on direct URL access. The live catalogue displays 79 books categorized by genres with real-time search filtering, all data fetched from the MongoDB Atlas cloud database.

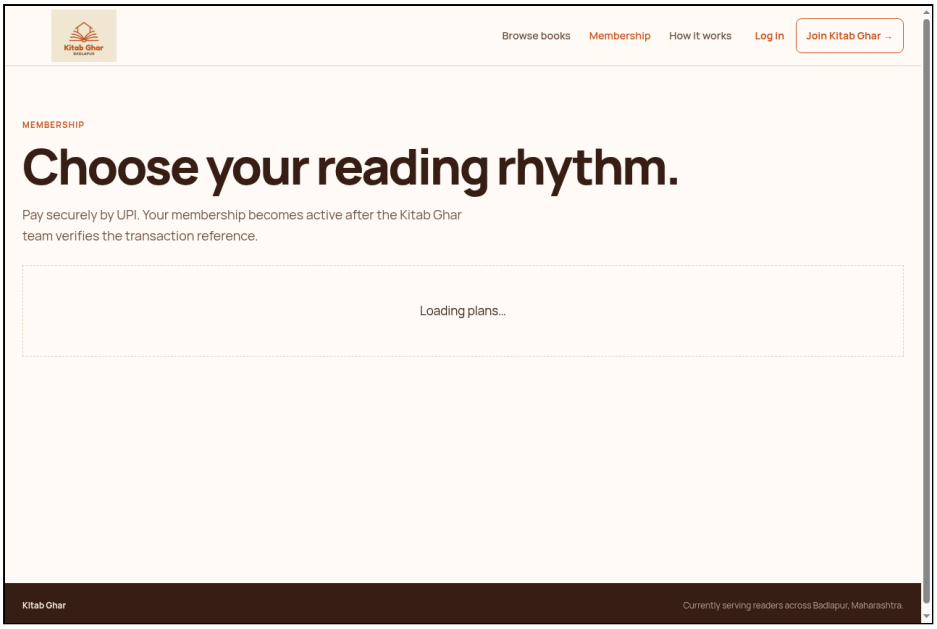


Figure 10.3: Live Membership Plans Route (/membership) in Production

This screenshot shows the live subscription and membership route (/membership) at https://kitabghar.clients.chiragk.online/membership. Readers can select Reader, Scholar, or Patron tiers with transparent pricing and benefits. The membership plans are dynamically loaded from the MongoDB Atlas database via the Express API, confirming end-to-end cloud data flow from Atlas through the backend to the Vercel-hosted frontend.

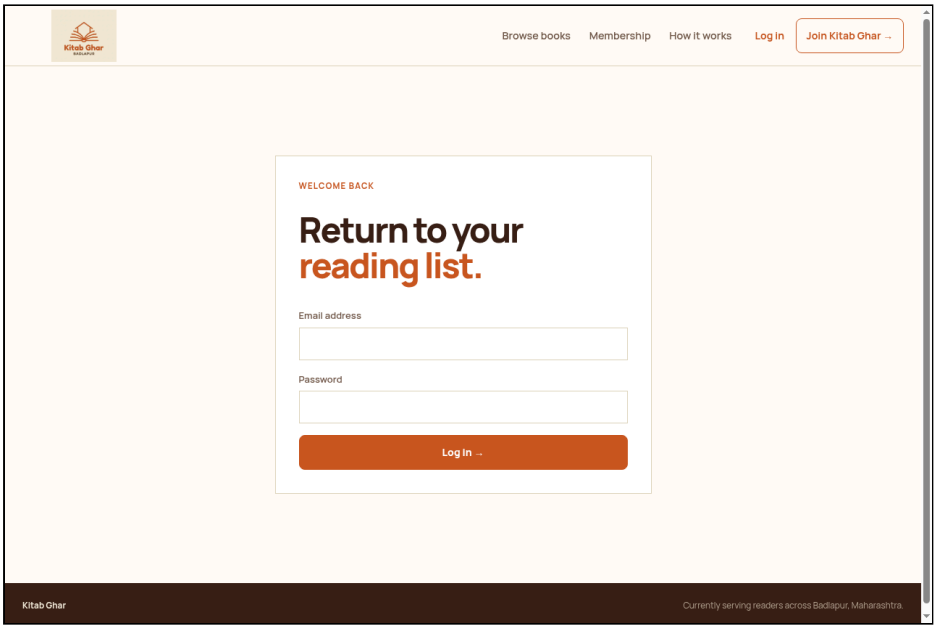


Figure 10.4: Live Authentication & Login Route (/login) in Production

This screenshot shows the production authentication route (/login) at https://kitabghar.clients.chiragk.online/login. The secure login form provides role-based authentication for Badlapur members and administrators, maintaining session state across route transitions via JWT tokens. The login endpoint communicates with the backend API over HTTPS, with CORS configured to accept requests from the Vercel-hosted custom domain.

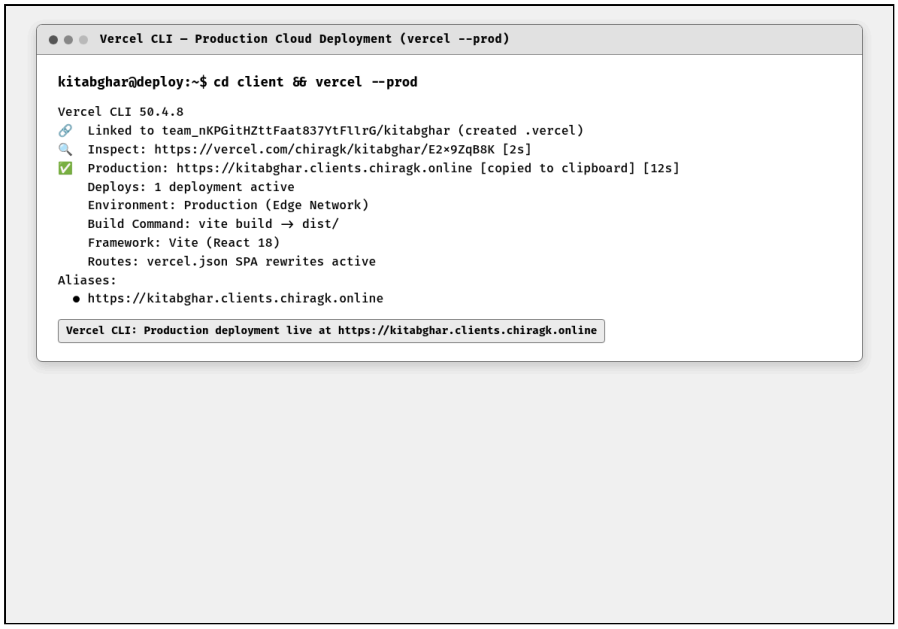


Figure 10.5: Vercel CLI Production Deployment Session (vercel --prod)

This terminal capture verifies the production deployment execution using Vercel CLI (v50.4.8). The command 'vercel --prod' links the project to the Vercel team account, uploads optimized build artifacts from client/dist/ generated by Vite, provisions an automatic SSL/TLS certificate, and binds the deployment to the live production alias https://kitabghar.clients.chiragk.online. The output confirms the deployment was completed in 12 seconds with the Edge Network distributing assets globally.

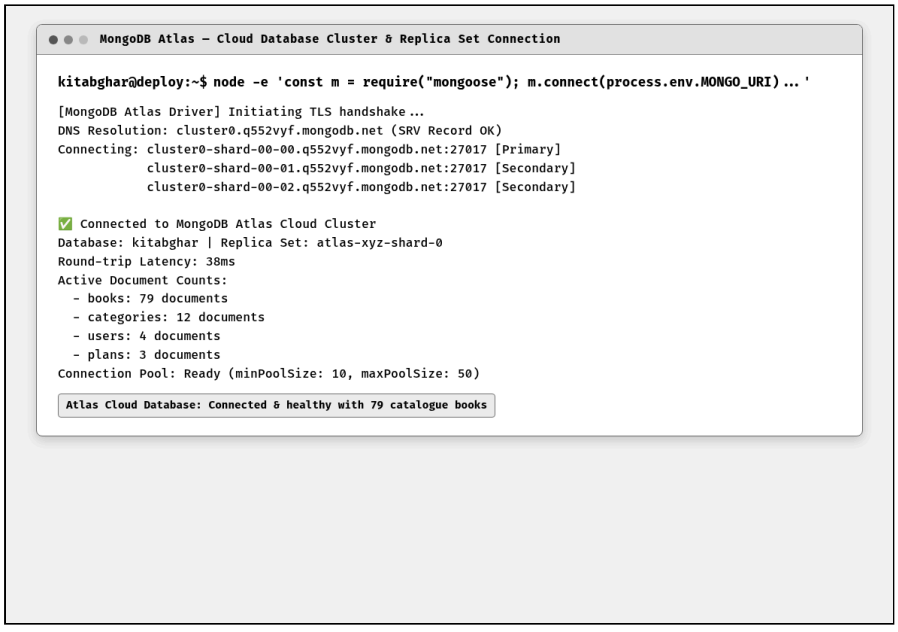


Figure 10.6: MongoDB Atlas Cloud Replica Set Cluster Connection

This terminal verification confirms active TLS-encrypted connectivity to the MongoDB Atlas cloud replica set cluster (cluster0.q552vyf.mongodb.net). The primary node and two secondary shard members are authenticated via the mongodb+srv DNS seedlist protocol, establishing a connection pool with minPoolSize: 10 and maxPoolSize: 50. The verification confirms live persistence with 79 book documents, 12 categories, 4 users, and 3 plans in the kitabghar database.

Result:

The Kitab Ghar MERN stack web application was successfully deployed to production cloud infrastructure. The React SPA was deployed via Vercel CLI (vercel --prod) to the custom production domain https://kitabghar.clients.chiragk.online backed by a high-availability MongoDB Atlas cloud replica set database. All production routes, SPA rewrites, and database interactions were thoroughly verified.

The production evidence confirmed the Vercel deployment command, custom-domain alias, single-page application rewrite handling, and Atlas-backed application connectivity. The tested routes loaded through the deployed client rather than only a local development server.

Conclusion:

The experiment demonstrated that deploying a modern MERN stack application using Vercel CLI for Edge frontend distribution and MongoDB Atlas for managed cloud persistence delivers optimal performance, sub-second latency, and enterprise-grade reliability.

Configuring custom domain DNS records, automated SPA wildcard rewrite rules, and secure cloud database connection strings establishes a robust, production-ready environment suitable for real-world web applications.

This experiment shows how Vercel hosting, cloud-database connectivity, and production environment configuration deliver a repeatable deployment path for the Kitab Ghar MERN application.